

Extraction of bioactive compounds

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Introduction

Bioactive compounds are distributed naturally in food and medicinal plants, and the effective extraction technique and proper purification protocol can ensure the therapeutic properties of the isolated bioactive compounds. Humans were using plants for the therapeutic purpose from the Greek and Roman period. But the knowledge of bioactive compounds present in herbal plants was missing in the ancient period. Later, the knowledge of bioactive compounds, isolation and purification procedure, and the physiological response of each bioactive compound were gained through tremendous research.

The history of extraction can be roughly the period when humans started using plants as medicine. Until the mid of 19th century, humans considered plants as their main source of therapeutic material, and the demand for medicinal plants is still increasing. In the ancient Indian medicinal system Ayurveda and Siddha, details of extracting the therapeutic compounds from the herbals were clearly mentioned. In 12th century, Acharya Shodhala described the process of “Arka Kalpana,” where the herbals/medicinal plants were soaked overnight in water, boiled the mixture and collected the condensed liquid for therapeutic treatment (Purnendu Panda et al., 2019). This Arka Kalpana procedure is similar to the so-called “distillation” technique. In the mid-18th century, the first Romanian Pharmacopoeia was written by a team of doctors and pharmacists with 301 monographs including herbal drugs (207 plants), animal origin (23 species), organic compounds (15), and

inorganic compounds (46). Interestingly, plenty of those herbal plants are still in use (Soroceanu, n.d.):

Matricaria chamomilia (common name: chamomile, antiinflammatory and antiseptic property),

- *Hypericum perforatum* (common name: St John's-wort),
- *Arnica Montana* (also known as: wolf's bane, antibiotics),
- *Chelidonium majus* (also known as: Greater celandine, improve eyesight).

Bioactive compounds are typically the secondary metabolites of the plant produced by following four major pathways: (1) Shikimic acid pathway, (2) malonic acid pathway, (3) Mevalonic acid pathway, and (4) nonmevalonate (MEP) pathway. Pharmaceutical/nutraceutical industries rely on different extraction techniques and many of those extraction techniques were in practice for the past 100 years (Azmir et al., 2013). Solvent extraction is widely used in industries and the extraction process includes the following stages: (1) penetration of solvent in the solid matrix (plant material); (2) solute (mixture of bioactive compounds and other phytochemicals) dissolves in the solvent; (3) diffusion of solute containing bioactive compounds out of the solid matrix; (4) collection of extracted solutes (Zhang et al., 2018). Isolation procedure begins with the extraction of bioactive compounds from the plants, however, some compounds are challenging to extract because of their insoluble nature. Further, the extractability of bioactives from the plant matrix depends on the compatibility of solvents and the suitable extraction technique.

The advances in extraction technology help in extracting valuable bioactive compounds from food waste including the waste generated from the food processing industries. Such waste generated during the industrial production setup alarmed the food scientists/process engineers. For instance, the United Nations Food and Agriculture Organization (FAO) estimated that around 1.3 billion metric tons of waste are generated every year from the food processing industries. Quantity of food waste is estimated to be one-third of the edible portion of food produced for human consumption (FAO, 2014; Parthasarathi & Anandharamakrishnan, 2016; Ramasamy & Subramanian, 2022). Interestingly, food waste contains plenty of nutritional compounds, and it is important to recover those compounds rather than landfilling or incinerating them. For instance, the fruit juice manufacturing industry will require only the flesh, and usually, the seeds and peels are considered as waste. But those seeds and fruit peels are loaded with phytochemicals and essential nutrients which can be isolated through proper extraction techniques (Sagar et al., 2018). For instance, apple processing industries produce the by-product "apple pomace" after pressing for juice, cider, distilled spirit, wine, and vinegar manufacturing. The

worldwide production of apple pomace is several metric tons every year and most of the apple pomace was used as feed for animals or disposed of as industrial waste. The solid residue of apple pomace is rich in valuable nutritional components including dietary fiber and polyphenolic compounds especially cinnamate esters, dihydrochalcones, and flavonols (Lyu et al., 2020). Fig. 2.1 illustrated the bioactive compounds present in the apple pomace.

There is a tremendous improvement in extraction technologies in the past five decades including the development of solvent (organic) less extraction techniques, reduce impurities, and low energy consumption. This chapter is aimed to describe the application of extraction techniques used in nutraceutical and pharmaceutical industries for the extraction of bioactive compounds.

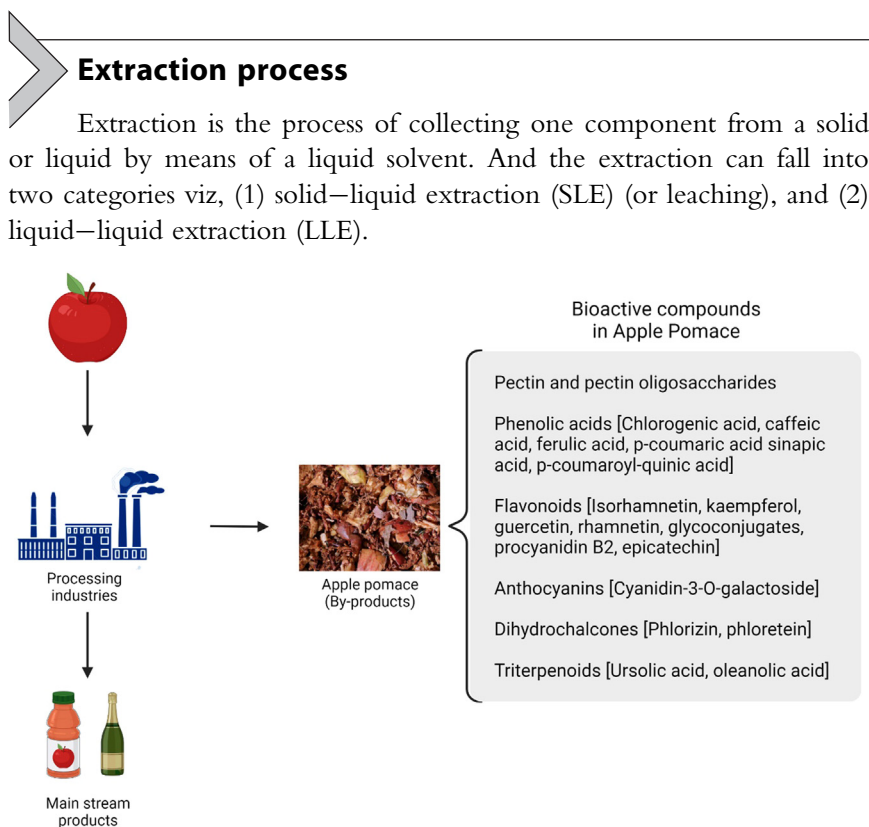


Figure 2.1 Bioactive compounds in the apple pomace. Created with www.Biorendor.com.

Solid–liquid extraction

SLE is the process of extracting soluble matter from an insoluble solid (plant materials) through solvents. SLE is also called lixiviation or leaching process. For instance, brewing coffee is a classic example of the SLE process, where the hot water (solvent) extracts the flavor components from the coffee powder (insoluble solid). Similarly, the extraction of curcumin from rhizomes of *Curcuma longa* by a suitable solvent is another classic example of the SLE process (Mandal et al., 2009). Fig. 2.2 illustrated the SLE process.

During SLE, the bound bioactive compounds from the solid matrices must be transferred either by dissolution or desorption into the solvent phase. Then the solvent mixture containing bioactive compounds can diffuse across the solid matrix by following stages (Chanoti et al., 2014; Siddique et al., 2020):

- Diffusion of solvent into the solid matrix,
- Dissolution of solute into the solvent phase,
- Diffusion of solvent–solute mixture (extract) to the surface of solid matrix,
- Transferring the extract from the surface into bulk solution and separation of extract from the solid matrix.

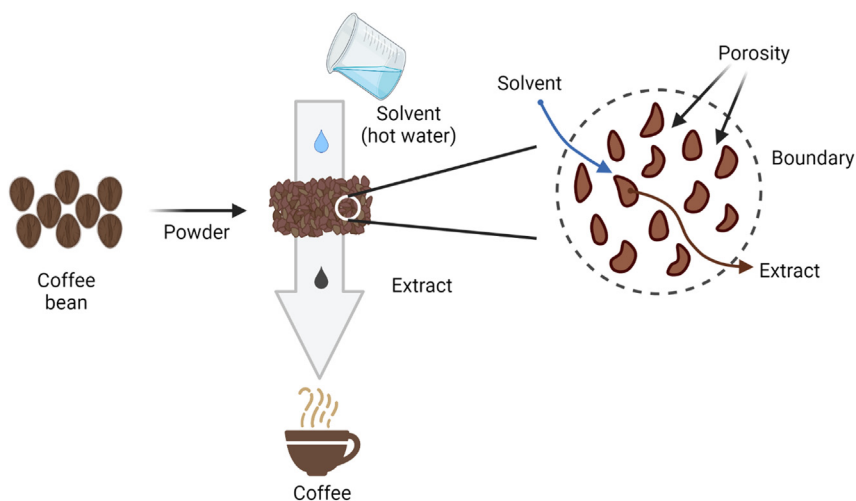


Figure 2.2 Extraction of solute (analyte) from the ground coffee beans. Created with www.Biorendor.com.

Liquid–liquid extraction

LLE is the process of dissolving a compound (analyte) from liquid mixture by contacting it with a suitable insoluble liquid solvent. For instance, the extraction of quercetin from a stock solution of Ginkgo biloba by contacting it with ethyl acetate. The stock solution and solvent forms a immiscible phase, and the quercetin will be dissolved by the organic solvent (Gamal et al., 2021). Industrially, the most preferred technique to extract an analyte from the liquid mixture is by simple distillation. However, distillation cannot be employed for the azeotropic mixtures (same boiling point for the liquid mixture) and for the heat-sensitive bioactive compounds. In such cases, LLE is employed for the extraction process. In this process, the extraction of bioactive compounds purely depends on solvent characteristics like miscibility, density, and solubility (Yahya et al., 2018). Isolation of bioactive compounds from the crude aqueous extract of herbal plants is governed by thermodynamic forces or electrochemical potential of ionic species. Generally, LLE process is carried out by following steps (Siddique et al., 2020):

- Mixing of two immiscible phases,
- Contact between feed and extraction solvent,
- Transfer of bioactive compounds from one phase to another,
- Recovery of bioactive from the solvent.

Based on the principles and technology, extraction techniques were classified as conventional techniques (Soxhlet extraction, decoction, maceration, hydrodistillation, and steam distillation) and nonconventional extraction technique (ultrasound-assisted, microwave-assisted, pulsed electric field extraction, supercritical fluid extraction (SFE), and enzyme-assisted extraction).



Conventional extraction techniques

Traditionally, our ancient people were treating diseases with the extracts of medicinal plants, and mostly aqueous extraction was conducted. And the principle of those extractions is still used in recent days with modifications. Conventional extraction techniques include maceration, percolation, decoction, hydrodistillation, reflux, and Soxhlet extraction (see Fig. 2.3). These conventional extraction techniques require either water or organic solvents as extracting medium and work at high temperatures. The efficacy of these extraction process is mainly determined by the solvent and are selected

Percolation	Flush with very hot solvent on a bed of finely divided plant materials. Example: Coffee brewing.
Decoction	Immersing plant materials in a boiling solvent. Example: Preparation of Indian chai
Infusion	Immersing plant materials in a hot solvent. Example: Preparation of British tea
Maceration	Plant materials are immersed in cold solvent. Example: Extraction of phenolic substances in red wine
Digestion	Hot maceration. Used in pharmaceutical field. Faster method and avoid contamination.
Elution	Separate solute from the surface of solid by solvent. Used in chemical analysis.

Figure 2.3 Conventional extraction techniques.

in accordance with the polarity of the bioactive compounds. The list of key solvents used for the extraction of bioactive compounds is illustrated in [Fig. 2.4](#). Other than solvent type, the following factors are also considered before the extraction process: molecular affinity between solvent and bioactive compound, mass transfer process, use of cosolvent mixture, human toxicity, and financial feasibility ([Gligor et al., 2019](#)).

These conventional techniques are preferred for their simplicity, easy installation, less operating cost, and less complication. The installation of microwave-assisted extraction technique requires more than \$3.5 million, while conventional extraction can be installed at less than \$1.5 million ([Fernández-Delgado et al., 2022](#)). This increasing installation and operating cost have a direct impact on the selling price of the bioactive compounds. [Table 2.1](#) described the advantages and limitations of conventional extraction technique. Examples of selected bioactive compounds extracted from conventional extraction techniques are listed in [Table 2.2](#).

Maceration

Maceration is one of the oldest and simplest extraction techniques, conducted at room temperature. In this technique, the coarse or powdered plant materials are soaked in solvent (water, oil, or alcohol) for longer time. Such prolonged soaking time ruptures the cell wall and drive the bioactives into the solvent. Then solvent can be strained off through a filter medium pressing the plant residues and recovering the bioactive compounds. The efficiency of bioactive

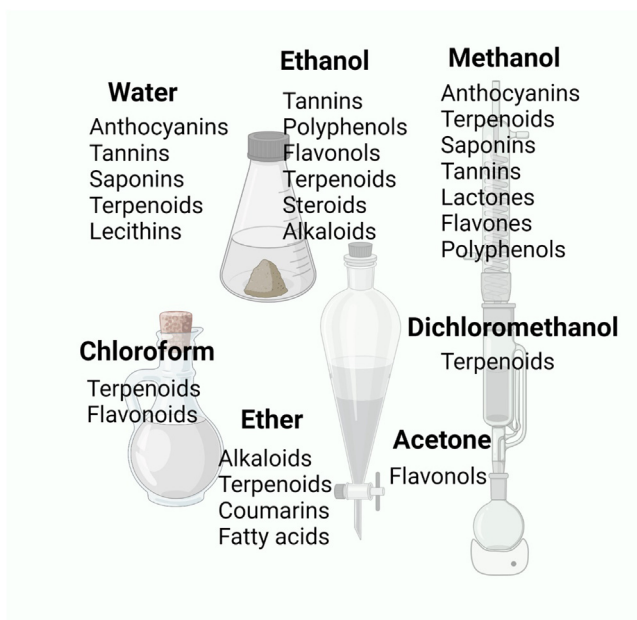


Figure 2.4 Solvents used for extraction for bioactive compounds. From Gligor, O., Mocan, A., Moldovan, C., Locatelli, M., Crişan, G., & Ferreira, I. C. F. R. (2019). Enzyme-assisted extractions of polyphenols – A comprehensive review. Trends in Food Science and Technology, 88, 302–315. <https://doi.org/10.1016/j.tifs.2019.03.029>.

extraction from the plants is determined by the solvent type, solvent polarity, plant material, agitation interval, and extraction time (Farooq et al., 2022). Agitation during the maceration process can improve the extraction yield by (1) promoting diffusion and (2) maintaining concentration gradient by separating concentrated solution from the sample surface and bringing fresh solution. The main advantage of this technique includes simple process, no heat involved, suitability for thermal sensitive material, low installation cost, and low maintenance cost. However, the drawback of maceration, such as long extraction time, poor yield, potential growth of spoilage organism, and possibility of fermentation, restricts the wide application of this technique in food sector (Jha & Sit, 2022). The main disadvantage of this conventional technique is the extraction time, which normally ranges from few hours to few days. In a recent study, Tewari et al. (2021) compared the extraction of three conventional techniques viz., maceration (in 96% methanol), infusion (in water) and Soxhlet extraction (in methanol 96%) evaluated the extracted bioactives on its biological activities. Aqueous extraction through infusion (45.04 mg GAE/g)

Table 2.1 Conventional extraction techniques: advantages and limitations.

Extraction technique	Advantages	Limitation
Maceration	Low investment cost, modulation of selectivity by solvent choice	Thermal destruction of some compounds
Hydrodistillation	Bioactive compounds separate automatically from the water	Some volatile components may be lost if the extraction temperature is too high
Infusion	It takes less time for extraction and latent heat of vaporization is less than water, so less heat will be consumed to make concentrated extract	Large amounts of solvent are required
Digestion	Extraction yield can be increased by using heating	It takes a long time
Decoction	Method of operation is simple	Large amounts of solvent are required
Percolation	It increases the extraction yield by providing better contact time between solvent and herbal material	It takes a long time and uses a lot of solvents
Soxhlet	Simplicity, applicability at high temperatures, boosts process kinetics	Poor extraction efficiency, time taking procedure

From Jha, A. K., & Sit, N. (2022). Extraction of bioactive compounds from plant materials using combination of various novel methods: A review. *Trends in Food Science and Technology*, 119, 579–591. <https://doi.org/10.1016/j.tifs.2021.11.019>.

showed the maximum highest phenolic content followed by Soxhlet (36.09 mg GAE/g) and maceration (29.43 mg GAE/g) technique. And maceration (12.55 mg RE/g) showed the highest flavonoid content followed by Soxhlet (10.65 mg RE/g) and infusion (4.86 mg RE/g) technique. The antioxidant activity of the extract by the DPPH method revealed maximum antioxidant potential in the order of aqueous infusion > Soxhlet > maceration.

Infusion

Infusion is similar to maceration, with a difference in the working temperature of the solvent and extraction time. Solvents like water or ethanol are most preferred for the extraction. For the extraction of phenolic

Table 2.2 Extraction of bioactive compounds from conventional extraction techniques.

Extraction technique	Plant material	Bioactive compound	Solvent	Research findings	References
Maceration	Wild apples	Total phenolic content, flavonoids, anthocyanins	Ethanol (70%, 96% v/v), propylene-glycol (45%, 80% w/w), olive oil, sunflower oil, and water	70% ethanol showed the maximum extraction of bioactive compounds.	Stojiljković et al. (2016)
Maceration	<i>Pistacia lentiscus</i> L. leaves	Identified 44 bioactive compounds including 2-furanmethanol, phenol, 2-carene, D-limonene, etc.	Macerate the plant material in methanol for 24 h	Macerated leaves showed 15–20 times higher in phenolic compounds and flavonoids with strong antimicrobial and antioxidant properties.	Djebari et al. (2021)
Maceration	<i>Rhamnus alaternus</i> leaves	Total polyphenols and flavonoids	Water: ethanol mixture and macerate for 12–24 h	Optimized extraction process using Box-Behnken RSM approach and determined the total polyphenols (230.9 mg GAE/g) and flavonoids (121.1 mg QE/g) content, with an IC ₅₀ value of 22.5 µg/mL.	Nekkaa et al. (2021)
Maceration	Olive leaves	Identified 36 polyphenols including chlorogenic acid, <i>trans</i> -ferulic acid, <i>trans</i> -caffeic acid, quercetin-3- <i>O</i> -rutinoside, oleuropein	75% ethanolic solution and macerate for 4 h	Extracted bioactive compounds were fortified with canola oil. The authors found maceration provided maximum extraction yield with TPA and antioxidant activity.	Dauber et al. (2022)

(Continued)

Table 2.2 (Continued)

Extraction technique	Plant material	Bioactive compound	Solvent	Research findings	References
Maceration	Huitlacoche (<i>Ustilago maydis</i>)	Total phenolic compounds	Ethanol (50%, 75%, and 99%)	Maximum total phenolic compounds were obtained for the maceration (13.94 mg GAE/mg) in 75% ethanol. For the same solvent, ultrasonication achieved only 7.28 mg GAE/mg of total phenols.	López-Martínez et al. (2022)
Maceration + Germination	<i>Sorghum bicolor</i> L.	Thiamine, riboflavin, pyridoxine, vitamin E, xanthophylls, flavonoids	Distilled water	Maceration improved the stability of bioactive compounds.	Pinheiro et al. (2021)
Maceration with heat, enzyme, microwave, and sonication	<i>Vitis vinifera</i> L. (grapes)	Anthocyanin, phenolic, and flavonoid contents	—	Microwave maceration and combination of microwave and sonication improved the anthocyanin, phenolic and flavonoid contents of the grape juices by approximately 100%, 20%, and 150%, respectively.	Guler (2023)
Infusion	<i>Achillea millefolium</i> , <i>Artemisia absinthium</i> , <i>Artemisia dracuncululus</i> , <i>Matricaria chamomilla</i> , <i>Salvia officinalis</i> , <i>Salvia pomifera</i> , <i>Salvia fruticose</i> , <i>Tilia × europaea</i> , <i>Valeriana officinalis</i>	Total Phenolic compounds	Distilled water (at 90°C) and infused for 10 min	<i>Salvia fruticose</i> showed the maximum TPC and highest antioxidant values.	Sotiropoulou et al. (2020)
Infusion	<i>Ilex paraguariensis</i> (Yerba mate)	Methylxanthines, phenolics, and saponins	Distilled water	Extraction pattern depends on the grinding process, and it determines the water permeability.	Kaltbach et al. (2022)

Hydrodistillation	Peppermint (<i>Mentha piperita</i> L.)	Essential oil containing menthol (35%–60%), menthone (2%–44%), and menthyl acetate (0.7%–23%)	Distilled water	Two hours of hydrodistillation possess significant amount of phenolic compounds, especially eriocitrin. Hydrosols after distillation showed higher antioxidant capacity and α -glucosidase inhibition capacity.	Berkas and Cam (2021)
Hydrodistillation	Italian garlic (<i>Allium sativum</i> L.) and elephant garlic (<i>Allium ampeloprasum</i> L.)	Volatile organosulfur compounds	Distilled water	Extraction followed by distillation and solid phase microextraction was compared for the effective extraction of garlic sulfides. Harsh thermal conditions during distillation resulted in higher quantity of volatile compounds, especially the tri- and tetrasulfide.	Ferioli et al. (2022)
Hydrodistillation	<i>Paeonia ostii</i>	Identified 150 volatile compounds including citronellol, D-Limonene, α -Pinene	Distilled water	Extraction showed highest amount of monoterpenoids (up to 77.28%), followed by alcohols (<26.82%) and aldehydes (<27.37%). And the extract hydrosols showed antibacterial activity against 8 Gram-positive and Gram-negative bacteria.	Xin et al. (2022)

(Continued)

Table 2.2 (Continued)

Extraction technique	Plant material	Bioactive compound	Solvent	Research findings	References
Hydrodistillation	<i>Pistacia atlantica</i>	Identified 95 bioactive compounds including α -pinene, β -citral, carvone hydrate, myristic acid, p-acetyltoluene, pinocarveol, and palustrol	Distilled water	Extracted essential oil showed excellent inhibitory effects on the growth of Gram-positive bacteria and less resistance to fungal samples (<i>Candida albicans</i>).	Hasheminya and Dehghannya (2020)
Ohmic hydrodistillation	<i>Achillea millefolium</i> L.	Identified 85 compounds including 1,8-cineole, chrysanthenyl acetate, α -pinene, terpinen-4-ol, cymene, etc.	Distilled water + salt to improve electrical conductivity	Essential oil from ohmic-assisted hydrodistillation had a better antimicrobial and antioxidant property than the hydrodistilled extract.	Karami et al. (2022)
Hydrodistillation + enzyme assisted	<i>Centella asiatica</i> (L.)	Asiaticoside, Total phenol content	400 mL of 0.1 M acetate buffer containing cellulase	Enzyme-assisted hydrodistillation in the presence of 10 FPU/g substrate extracted more phenolic compounds in both essential oil (107.05 ± 0.30 mg GAE/100 g) and hydrosols (23.11 ± 0.43 mg GAE/100 g).	Abu Bakar et al. (2022)
Hydrodistillation + pressurized solvent extraction	<i>Pistacia lentiscus leaves</i>	Quercitrin and loliolid	Water (for distillation). Methanol (for pressurized solvent extraction)	Isolated bioactive compounds induce the enzyme inhibition and quercitrin attenuates the lysozyme fibrillation at 25 ug/mL.	Maalej et al. (2021)

Sonohydro-distillation; microwave-assisted hydrodistillation	<i>Cinnamomum cassia</i> barks	Identified 11 compounds including Camphene, o-cymene, gamma-terpinene, piperonal, etc.	Water	Microwave-assisted technique had rapid extraction kinetics with less energy consumption and lower solvent requirement. Microwave and sonohydrodistillation showed higher recovery of bioactive compounds than the conventional distillation.	Benmoussa et al. (2023)
Soxhlet extraction	Orange peel	Phenolic compounds	de-ionized water	The authors encapsulated the orange peel extract using WPI and found the total phenolic compounds didn't affect even during 150°C of spray drying.	Edrisi Sormoli and Langrish (2016)
Soxhlet extraction	Sea buckthorn berries	β -carotene, α -tocopherol	Ethanol, 2-MTHF, Hexane, diethyl ether	Soxhlet extraction method had significantly higher sea buckthorn oil than the accelerated solvent extraction technique. Authors optimized the extraction temperature of 120°C for hexane and diethyl ether, 90°C for ethanol, and 150°C for 2-MTHF for the maximum extraction yield.	Vilas-Franquesa et al. (2022)
Soxhlet extraction	<i>Azadirachta indica</i>	Tannins, amino acids, alkaloids, gallic acid, flavonoids, etc.	Ethanol (95% v/v)	In vivo studies with rabbits showed significant reduction in total cholesterol, triglycerides, and low-density lipoproteins.	Asghar et al. (2023)

compounds from herbals, polar solvents derive higher total phenolic content compared to the less polar solvents (Sotiropoulou et al., 2020). Some of the popular infusion practice includes infusion of nutrients from tea (Karak et al., 2017), infusion of aromatic hydrocarbons from Yerba Mate (Panzl et al., 2022), and bioactive compounds from flower petals (Pires et al., 2017).

Soxhlet extraction

Soxhlet extraction is a conventional and widely used technique in the pharmaceutical sector for the extraction of bioactive compounds from plant materials. The choice for Soxhlet extraction is mainly due to the simple operation procedure with low equipment cost, efficient recovery of bioactives, and comparatively less extraction time than the other conventional methods. This classical extraction apparatus was first designed by Franz von Soxhlet in 1879, and surprisingly the developed Soxhlet apparatus was in practice until today in many research labs. The survival of Soxhlet apparatus till now is also attributed to the acceptance of this technique by many referencing agencies including/referencing agencies including USFDA (United States Food and Drug Administration), the United States Environmental Protection Agency, the Official method of analysis of AOAC international, and British Standards (Barroso et al., 2013).

Briefly, the lab scale conventional Soxhlet extraction instrument consists of three units including round bottom flask (RBF), extractor, and condenser as shown in Fig. 2.5. Initially, the RBF in the Soxhlet extractor is loaded with solvent, and thimble in the extractor is loaded with plant materials (particulate form). When the solvent reaches its boiling point, the vapor travels to the condenser, and the condensed hot solvent falls on the thimble and extracts the bioactive compounds. Once the condensed solvent reaches the overflow level in extractor, siphon aspirates it to the RBF. And the same process is repeated until the complete extraction is achieved, which can be visually inspected by the colorless form of solvent that aspirates to the RBF. Then the solvent-bioactive mixture can be evaporated by the rotary evaporator.

The Soxhlet apparatus was in use for a long time and employed as a standard technique to perform SLE. And the extraction should be optimized for the type of solvent, temperature, solvent-to-solid ratio, extraction time, and particle size to achieve maximum yield.

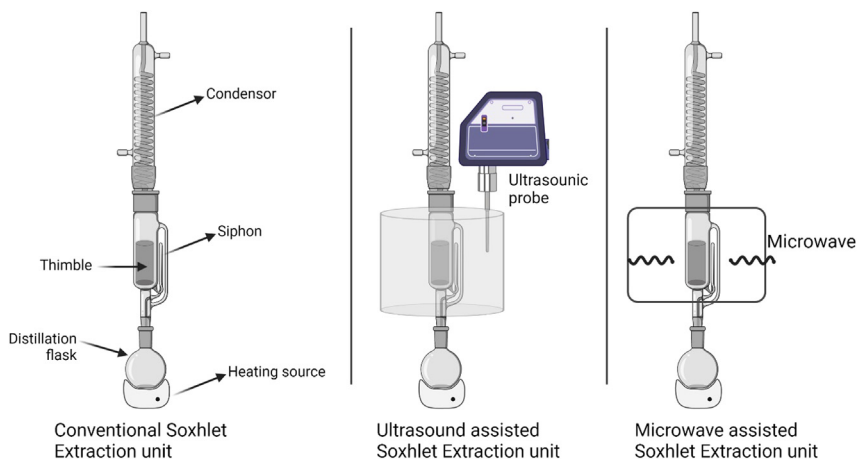


Figure 2.5 Conventional and second-generation Soxhlet extraction apparatus. Created with Biorender.com.

Working solvent: Suitable solvent should be selected for the extraction process as the extraction efficiency, environmental impact, application, and renewability differs for each solvent. Some of the commonly used for extraction include acetone, ethanol, diethyl ether, petroleum ether, *n*-heptane, isopropanol, chloroform, methanol, and 1-butanol. *n*-Hexane is widely used for its excellent oil solubility, ease of recovery, and for narrow boiling point (63°C –69°C) (Keneni et al., 2021).

Particle size: The plant material or seeds loaded in the thimble should be pulverized to improve extraction efficiency. Reducing the particle size can improve the surface area enormously and improve the solvent accessibility to extract. However, very fine particles may also reduce the oil extraction efficiency, as the oil may stick together with the fine particles and forms a paste-like consistency and prevent interaction between the solvent and particles. A study with Jatorpa seeds resulted in a maximum yield of 46% for intermediate-sized particles (0.5–0.75 mm), 43.3% for largest particles (> 0.75 mm), and least yield of 40% for the fine particles (<0.5 mm) (Sayyar et al., 2013).

Solvent-to-solid ratio: The major driving force for the extraction process is the concentration gradient between the solid and the solvent. Thus, the solvent-to-solid ratio provides a positive impact on the extraction yield, that is, increasing the solvent-to-solid ratio increases the yield (Pinelo et al., 2005).

Second-generation Soxhlet extraction

The major drawback of the Soxhlet apparatus is the long extraction time and requirement of large quantity of solvent. Such generation of solvent at the end of extraction is expensive to dispose of and it can also create an environmental impact. One way to reduce the extraction time is by improving the mass transfer through agitation. Thus, the second generation of Soxhlet apparatus is developed with the aim to reduce the extraction time. See [Fig. 2.5](#) for the schematic representation of second-generation Soxhlet extraction unit.

Ultrasound-assisted Soxhlet extractor: [Luque-García and Luque de Castro \(2004b\)](#) first accelerated the Soxhlet extraction process by combining the ultrasonication technique. The authors found that the ultrasound can effectively reduce the number of Soxhlet extraction cycles, and the quantitative values of extraction were similar to the conventional Soxhlet extraction and ISO reference methods. Later, [Djenni et al. \(2013\)](#) modified the Soxhlet apparatus by directly inserting the ultrasonic probe and found that the ultrasound-assisted Soxhlet extraction reduced the extraction time by a factor of 16 compared to the conventional technique. However, only limited work on incorporating ultrasound with Soxhlet was reported, which might be due to the difficult in modifying the Soxhlet apparatus and other safety issues in working with the organic solvents.

Microwave-assisted Soxhlet extractor: The same group [Luque-García and Luque de Castro](#) who developed the ultrasound-assisted Soxhlet extractor, designed the Soxhlet with microwave generator. The idea is to accelerate the extraction by passing the microwaves through the Soxhlet apparatus ([Luque-García & Luque de Castro, 2004a](#)).

Distillation

Soxhlet extraction may not be a suitable method for volatile bioactive compounds. In such cases, extraction with water or steam (through distillation) is a preferred alternative for extracting volatile oil and essential oils from plants. The distillation process is simple, where steam is passed through the plant material of interest and condenses the steam. The condensed steam contains volatile oils or essential oils, which are separated by phase separation. For industrial applications, three types of distillation were employed: steam distillation, hydrodistillation, and dry steam distillation (see [Fig. 2.6](#)).

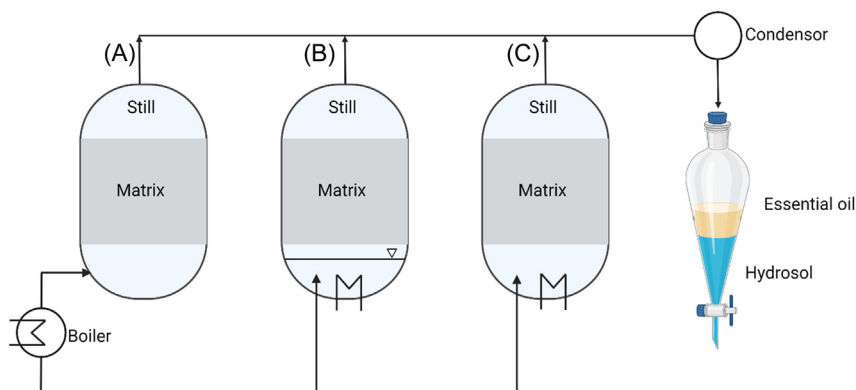


Figure 2.6 Generalized flow of different steam distillations: (A) dry steam distillation, (B) direct steam distillation, and (C) hydrodistillation. *Created with Biorender.com.*

In steam distillation (also known as direct steam distillation), where the plant material is placed over the perforated grid (not in direct contact with the water), steam is generated from the bottom of the still. In the hydrodistillation, the plant material is immersed in boiling water, and the generated steam will be condensed. Whereas in dry steam distillation, the steam is generated outside the still and is passed through the perforated grid. The main difference between steam distillation and dry steam distillation is where the steam is generated (Prado et al., 2021). The advantages and disadvantages of steam distillation are given in Box 2.1.

Fagbemi et al. (2021) compared the extraction of volatile compounds using Soxhlet extraction and hydrodistillation and found different compositions of essential oil. The authors found higher yield for Soxhlet extract (2.5%) than the hydrodistillation technique. The selective yield increase for the Soxhlet is mainly attributed to the longer extraction time (6 hours) compared to the distillation (3 hours). The increase in extraction time allows the sufficient degradation of plant tissues with adequate release of volatile bioactive compounds. Although hydrodistillation resulted in lower yields however it produced higher-quality oxygenated compounds indicating the minimized thermal decomposition of oxygenated compounds. The detailed comparison of oxygenated compounds obtained by Soxhlet extraction and distillation is given in Table 2.3.

Box 2.1 Advantages and disadvantages of steam distillation.

Advantages:

- Inexpensive equipment
- Can be utilized for industrial scale
- No need for subsequent separation step
- Organic solvent-free technique

Disadvantages:

- Thermal sensitive bioactive compounds may degrade
- Long extraction time
- High energy consumption

Data from Prado, J. M., Vardanega, R., Debien, I. C. N., Meireles, M. A. A., Gerschenson, L. N., Sowbhagya, H. B., Chemat, S., Prado, J. M., Vardanega, R., Debien, I. C. N., Meireles, M. A. A., Gerschenson, L. N., Sowbhagya, H. B., & Chemat, S. (2021). *Conventional Extraction* (pp. 109–127). Elsevier B.V. <https://doi.org/10.1016/b978-0-12-820563-1.00015-9>.

Table 2.3 Comparison of terpenoids extracted from hydrodistillation and Soxhlet extraction technique.

Terpenoids	Compound	Hydrodistillation	Soxhlet extraction
Monoterpene hydrocarbons	p-Xylene	0.75	0.08
	o-Xylene	0.77	—
	1-Decyne	—	0.2
	Benzene, 1-methyl-3-propyl	1.55	—
	Naphthalene, 2-methyl-	0.14	—
	Naphthalene, decahydro-, trans	1.75	—
Oxygenated monoterpenes	Benzeneethanol, alpha-, beta-dimethyl-	—	0.14
	2,4-Decadienal, (E, E)-	0.16	—
	Cyclohexane, (1-methyl propyl)-	1.09	—
Oxygenated sesquiterpenes	Oxalic acid, isobutyl nonyl ester	—	0.14
Terpenes	2-Methyltetracosane	9.41	1.15
Oxygenated diterpenes	1-Eicosanol	0.63	—

Source: Data from Fagbemi, K. O., Aina, D. A., & Olajuyigbe, O. O. (2021). Soxhlet Extraction versus Hydrodistillation Using the Clevenger Apparatus: A Comparative Study on the Extraction of a Volatile Compound from *Tamarindus indica* Seeds. *Scientific World Journal*, 2021. <https://doi.org/10.1155/2021/5961586>.



Nonconventional extraction techniques

Although conventional extraction techniques are widely used in industries, the main drawbacks associated with these conventional techniques such as large solvent requirement, use of organic solvent, long extraction time with less selectivity, and poor yields. Further, the increasing awareness of green chemistry made the researchers look for alternative techniques. The concept of green chemistry was first proposed by Prof. Anastas and Prof. Warner in 1998, defining “utilization of a set of principles that reduces or eliminates the use or generation of hazardous substances in the design, manufacture, and application of chemical products” (Anastas & Warner, 1998). The application of green chemistry concepts in the extraction sector (green extraction) can potentially reduce the extraction time and application of hazardous organic solvents. Green extraction is defined as the “design of extraction process will reduce energy consumption, allows the use of alternative solvents and renewable natural product, and ensure a safe and high-quality extract/product” (Chemat et al., 2012). Briefly, the concept of green extraction encourages to use: (1) renewable plant resources, (2) alternative solvents (water), and (3) innovative technologies to reduce energy consumption. It also encourages to: (1) generate by-products instead of waste, (2) reduce unit operations, and (3) produce nondenatured and biodegradable extracts without contaminants (Chemat et al., 2012). Based on these, novel extraction techniques such as enzyme-assisted extraction, pulsed electric field, ultrasound-assisted extraction (UAE), microwave-assisted extraction, and SFE are quite popular in food sector. Examples of selected bioactive compounds extracted from nonconventional extraction techniques are listed in Table 2.4.

Microwave-assisted extraction

Microwave is a nonionizing type of electromagnetic energy with a frequency and wavelength between 300 MHz to 300 GHz and 1 m to 0.1 cm, respectively. Most of the available microwave instruments (both industrial and domestic instruments) operates at a fixed frequency of 2.45 GHz. The extraction process of microwave technique is quite different from the conventional technique, where the microwave heat utilizes the energy of microwave to heat the material through ionic conduction and dipole rotation. Thus, instant heating resulted from the simultaneous

Table 2.4 Extraction of bioactive compounds from novel extraction techniques.

Extraction technique	Plant material	Bioactive compound	Solvent	Condition	Concentration	Research findings	References
Ultrasound-assisted extraction (UAE)	<i>Psidium guajava</i>	Polyphenols, flavonoids, vitamin C	Water	62.19°C; 14.94 min	TPC—68.8 mg GAE/g; TFC—289.77 mg QE/g; Antioxidant—80.3%; Vitamin C—25.7 mg/100 g	Pulsed UAE reported better extraction than the normal sonication method.	Wani and Uppaluri (2022)
	Apple pomace	Polyphenols and flavonoids	Natural deep eutectic solvents (NDES)	20 kHz, 400 W, and 70% amplitude; 83.2 W/cm ² acoustic intensity	TPC—5.6 mg GAE/g; TFC—5.4 mg QE/g	Authors found three deep eutectic solvents (Choline chloride with glycerol or lactic acid or citric acid) for maximum extraction efficiency.	Rashid et al. (2023)
	Chokeberry	Anthocyanin	NDES	42.7°C, 90 min, and 40% (w/w) water in NaDES	5.99 ± 0.23 mg CGE/g DW (TAC)	Authors revealed cyanidin molecule from chokeberry interact with NDES constituents by carbon—hydrogen bonds and cation- π interactions.	Jovanović et al. (2023)
	Sea buckthorn	Identified 24 bioactive compounds including phytosterols, monoterpenoids, PUFA, MUFA, and SFA.	<i>n</i> -Hexane	Optimal condition 50°C for 8.28 min at 552 W	Carotenoid: 237.04 ± 3.86 mg per 100 g β -carotene equivalent	Carotenoid content increases for the shorter extraction condition.	Sanwal et al. (2022)

Microwave-assisted extraction (MAE)	<i>Citrus sinensis</i> peels	Identified gallic acid, chlorogenic acid, caffeic acid, ferulic acid, <i>p</i> -coumaric acid, catechin, rutin and uercitin	Acetone + water	51% acetone concentration in water (v/v); 500 W; 122 s; 25 mL/g solvent-to-solid ratio	TPC—12.20 mg GAE/ g	MAE provided higher recovery of TPC along with high-quality phenolic compounds than the UAE, Soxhlet and conventional extraction techniques.	Nayak et al. (2015)
	Salmon side streams	DHA and EPA	<i>n</i> -Hexane	14.6 min and 291.9 W for Salmon backbones, 10.8 min for 50.0 W Salmon heads, and 14.3 min, 960.6 W for viscera.	Salmon oil: 39.4 g/100 g dw for backbones, 38.1 g/100 g dw for salmon heads, and 71 g/100 g dw for salmon viscera.	MAE recovered 69% of total fat in salmon backbones and heads, and 92% in salmon viscera in less than 15 min of extraction.	de la Fuente et al. (2022)
	<i>Pistacia vera</i> L. leaves	Phenolic compounds and flavonoids	EtOH:H ₂ O (70:30, v/v)	Varied the extraction time and temperature between 2 to 20 min and 50°C to 70°C	TPC—196.35 mg GAE/ g; TFC -83.34 mg CE/g	Extracts of microwave showed better biological activity (antioxidant and antimicrobial activity) than the extract of conventional maceration.	Elakremi et al. (2022)
	<i>Jackfruit pulp (Artocarpus heterophyllus Lam)</i>	Phenolic compounds	Water; 60% methanol; 60% ethanol; 60% acetone	Extraction for 165 s at 550 W	TPC—1.11 mg GAE/g (water extract); 1.43 mg GAE/g (60% methanol extract); 2.40 mg GAE/g (60% ethanol extract) and 1.49 mg GAE/g (60% acetone extract)	Phenolic content varied significantly for the solvent and extraction technique. MAE ethanol extract had highest antioxidant activity than the extracts of ultrasound and ultrasound-microwave assisted extraction.	Cheng et al. (2023)

(Continued)

Table 2.4 (Continued)

Extraction technique	Plant material	Bioactive compound	Solvent	Condition	Concentration	Research findings	References
Microwave + ultrasound-assisted extraction	Tomato seed	Identified linoleic acid, stearic acid and oleic acid in the extracted tomato seed oil	Ethanol: Ethyl acetate (2:3)	40°C, 540 W (microwave), 60 W, and 40 min (ultrasound)	Linoleic acid—55.3%	Dipole rotation and dielectric effect (microwave) and mechanoacoustic effect (ultrasound) improved mass transfer of tomato seed oil extraction.	Kumar et al. (2023)
Supercritical fluid extraction (SFE)	<i>Curcuma longa</i> and <i>Curcuma amada</i> (Turmeric)	Oleoresin and curcuminoids	SC-CO ₂ and cosolvents like methanol, ethanol, and acetone at 10%–30% v/w.	100 to 400 bar, 40 °C, 2 L/min, and 60 min	Oleoresin— 35.37 ± 3.43 mg/g; curcuminoids— 276.95 ± 7.89 µg/g	Use of ethanol as cosolvent improved the extraction efficiency than the other cosolvents. Enzymatic pretreatment of plant material further improves the extraction efficiency.	Nagavekar and Singhal (2019)
	<i>Phoenix dactylifera</i> (Date seeds)	Phenolic compounds, flavonoids, anthocyanins, and carotenoids	SC-CO ₂ and Ethanol as cosolvent.	Temp: 35°C to 70 °C; Operating pressure: 15–40 MPa; CO ₂ flow rate: 0.5–5 mL/min.	TPC—274.98 mg GAE/100 g; flavonoids—141.78 mg QE/100 g; anthocyanins—1.00 mg/100 g; carotenoids—1.91 mg BCE/100 g	Supercritical and subcritical extraction techniques showed maximum yield with rich phenolic, flavonoid, anthocyanin, and carotenoid contents than the extract of Soxhlet technique.	Ghafoor et al. (2022)
	Used coffee grounds	Caffeine, DUFA, and MUFA	SC-CO ₂ and cosolvents like isopropanol, ethanol, and ethyl lactate	Temp: 313 to 333 K; 20 to 50 MPa; 1.9×10^{-3} kg/min	Caffeine content 3.96%; DUFA 42% and MUFA 15.8%	Fatty acid analysis of extract revealed major portion of palmitic and linoleic esters. SFE with cosolvents improved the antioxidant activity by 12.5 times.	Coelho et al. (2020)

HHP + SFE	<i>Physalis peruviana</i> L. pulp (Cape gooseberry)	phenolics and β -carotene	SC-CO ₂ and ethanol	HHP condition: 300 to 500 MPa and 1–3 min of holding time. SFE condition: 25 MPa; 313.15 K with a CO ₂ flow rate of 1 mL/min	β -carotene: 5.51 mg /100 g d.m TPC: 29.28 mg GAE/100 g d.m	Combination of HPP and SFE enhances the extraction efficiency of bioactive compounds. however, during 60 days storage extracts underwent carotenoid degradation due to geometric isomerization and oxidation.	Torres- Ossandón et al. (2018)
Enzyme assisted Extraction	<i>Prunus avium</i> L (sweet cherry) pomace	Phenolic compounds and proanthocyanidins	Enzymes: Depol 740 L Promod 439 L and Pectinase 62 L	Temp: 30°C–70 °C; Time: 30 –300 min; enzyme conc: 1–120 μ L/g; pH: 3.0–8.0	TPC: 1.87 mg GAE/g sample;	Acid hydrolysis showed maximum phenolic content followed by Promod enzyme and alkaline hydrolysis.	Dominguez- Rodríguez et al. (2021)
	<i>Sargassum boveanum</i> , <i>Sargassum angustifolium</i> , and <i>Feldmannia irregularis</i> (Brown algae)	Phenolic compounds	Enzymes: Carbohydrases (Viscozyme L, Celluclast 1.5 L FG, Termamyl 120 L, Ultraflo L) Proteases (Flavourzyme, Alcalase, Neutrase)	Optimal condition: 1 mL of 0.1% enzyme, 20 h extraction time in 350 rpm stirring	84.0 mg GAE/g sample	Enzymes breakdown the algae cell wall and release active compounds into medium.	K. Habeebullah et al. (2020)
	Raspberry pomace	Essential fatty acids, tocols, phytosterols, and ellagitannins	Cellulase, pectinase, protease	Temp: 45°C, Time 1 hr	Polyphenols: 3.7 g/100 g Total ellagic acid: 1.46 g/100 g	Antioxidant activity of enzyme-assisted extracts was 25% higher than the organic solvent extract.	Saad et al. (2019)

TPC, Total phenolic compounds; TFC, total flavonoid compounds; TAC, total anthocyanin content; PUFA, polyunsaturated acid; MUFA, monounsaturated fatty acid; SFA, saturated fatty acid; DHA, docosahexaenoic acid; EPA, eicosapentaenoic acid; GAE, gallic acid equivalent; CE, catechin equivalent.

occurrence of ionic conduction and dipole rotation is absorbed directly by the medium (solvent). But the main drawback with the microwave technique is the solvent with no dipole moment (like nonpolar solvents) cannot be directly heated by the microwaves. Thus, microwave heating is applied only to the selected and targeted materials based on their dielectric constants, and the efficiency of microwave heating depends on the dissipation factor of the material ($\tan \delta$) as given by the following equation (Chan et al., 2011):

$$\tan \delta = \frac{\varepsilon''}{\varepsilon'} \quad (2.1)$$

where ε'' is the dielectric loss indicating the efficiency of converting microwave energy into heat and ε' is the dielectric constant which measures the ability of the material to absorb microwave energy. In microwave-assisted extraction, the solvent with higher ε' is chosen for the maximum yield. The dielectric constant and dielectric loss constant of selected solvents are given in Table 2.5.

And the absorbed electrical energy is converted into thermal energy by the following equation (Chan et al., 2011):

$$P = K \cdot f \cdot \varepsilon' E s d^{23} \pi r^2 \quad (2.2)$$

where P is the microwave power dissipation per unit volume, K is the constant, f is the applied frequency, E is the electric field strength and $\tan \delta$ is the dielectric loss tangent. Thus, microwave-assisted extraction can complete the extraction in shorter time and accelerate the extraction process by rapid heating of sample-solvent mixture (Parthasarathi et al., 2013).

Table 2.5 The dielectric constants and dielectric loss constants of six common solvents (CEM, n.d.).

Solvent	Dielectric constant (ε')	Dielectric loss constant (ε'')
Water	80.4	9.889
DMSO	45.0	37.125
Acetonitrile	37.5	2.325
DCM	9.1	0.382
Toluene	2.4	0.096
Hexanes	1.9	0.038

Ultrasound-assisted extraction

UAE is a novel nonthermal food processing technique applied in various food processing industries including thermal treatments, extraction, alteration of textural behavior, and food preservation. Compared to conventional techniques, UAE is preferred for its rapid extraction time, instrument simplicity, reduced power consumption, and reduced consumption of solvent. Ultrasound waves are typically in the frequency ranges of 20 kHz to 10 MHz, and [Picó \(2013\)](#) classified ultrasound based on its intensity as low-intensity ultrasound and high-intensity ultrasound. Low-intensity ultrasounds are low power (less than 1 W/cm^2) with high frequency (100 kHz to 1 MHz), usually applied for noninvasive analysis of food including the composition of meat products. High-intensity ultrasounds are high power ($10\text{--}1000 \text{ W/cm}^2$) with low-frequency range (16–100 kHz), used for improving functional characteristics like emulsification, defoaming and solubility of protein, inactivating microorganisms, and extraction of bioactive compounds ([Arvanitoyannis et al., 2017](#); [Picó, 2013](#)).

The novelty of ultrasonication is the generation of acoustic cavitation, which consists of rapid growth and collapse of microbubbles in water (see [Fig. 2.7](#)). For instance, when sonicating a liquid medium at high ultrasound intensity, then the sound waves propagate into the liquid media and result in alternating high-pressure (compression) and low-pressure (rarefaction) cycles. Small vacuum bubbles or voids in the liquid are generated during the low-pressure cycle, and in the high-pressure cycle collapse the generated bubble. The subsequent generation of void bubbles and immediate collapse is also called as cavitation phenomenon, which

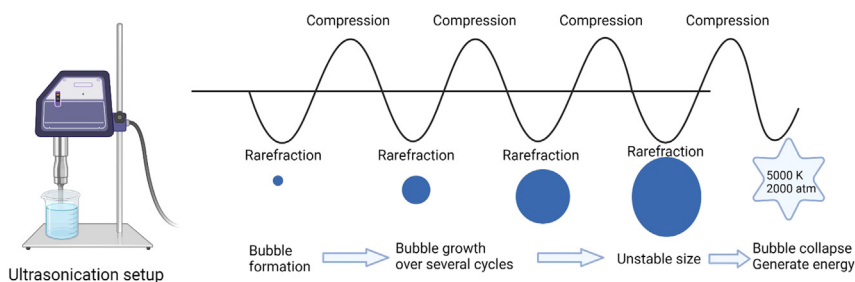


Figure 2.7 Ultrasonication generation of acoustic cavitation during ultrasonication. Created with Biorender.com.

can generate high temperatures (5000 K) and pressures (2000 atm) (Heilscher, 2022).

The mechanism behind UAE is the application of acoustic cavitation near the surface of plant cell wall and exert mechanical and thermal effects, which leads to cell wall disruption and subsequently improves the penetration of solvent and diffusion of cell material. Thus the performance of UAE depends on ultrasonic power, frequency, temperature, ultrasonication time, solvent properties, and material to solvent ratio (Martínez-Patiño et al., 2019). There are six different mechanisms involved in the extraction of bioactive compounds and the combination of six mechanisms helps in improving the yield of extraction (Chemat, Rombaut, Sicaire, et al., 2017; Khadhraoui et al., 2018).

Fragmentation: Inter-particle collision and shockwaves lead to fragmentation with a drastic reduction in particles size. Consequently, there will be an increase in surface area of the plant matrix and increased mass transfer of bioactives compounds with maximum yield.

Erosion: Implosion of cavitation bubbles near the surface of plant matrix induces erosion in plant structures and transfer the bioactives to liquid medium. This mechanism enhances the accessibility of solvent to the plant matrix and improved the extraction yield.

Capillarity and sonoporation: Capillarity (also known as ultrasonic capillary effect) relates the depth and velocity of penetration of ultrasonic waves with the extraction. Sonoporation refers to the formation of tiny holes in the cell membrane to transfer the cell materials to the outer medium.

Local shear stress: During ultrasonication, high shear forces and turbulences are generated within the fluid by the collapse of cavitation bubble, which ruptures the plant matrix and improves the extraction yield.

Detexturation: The generated shear force results in deformation of surface in the abaxial and adaxial surface of the material with a surface microfracture. This detexturation enhanced the accessibility of solvent and penetrates the plant matrix.

Supercritical fluid extraction

SFE is an innovative extraction technique, gaining importance in the past two decades. The extraction process in the SFE is allowing the solvent to contact the plant matrix and collects the compound of interest from the plant matrix (Mezzomo et al., 2009). But the novelty of this technique is

the application of pressurized solvent for extraction technique. SFE is an ideal technique for the extraction of essential oils, carotenoids, flavonoids, and fatty acids from plant matrix and is employed as a successful alternative to conventional extraction techniques. Memon et al. (2016) compared the conventional extraction and SFE for the recovery of flavanones, chalcone, and triterpenoids from *Syzygium campanulatum* Korth leaves. Three different conventional extraction methods including maceration (MAC), reflux (REF), and soxhlation were employed with ten different solvents of water, 0.1 mol/L HCl, 0.1 mol/L NaOH (at 95°C–100°C), H₂O:EtOH (1:1, v/v), H₂O:MeOH (1:1, v/v), EtOH, MeOH, EA, *n*-HEX, and *n*-HEX:MeOH (1:1), at 65°C–80°C for 24 hours. The authors found SFE method with ethanol cosolvent showed the maximum recovery of 84.9% compared to the conventional extraction technique (66%).

Supercritical fluid is a compressed gas that has a density similar to that of liquid. Fig. 2.8 shows the phase diagram of a material. At triple point (TP), three phases of the substance (gas, liquid, and solid) coexist in thermodynamic equilibrium. Further increasing the temperature, the substance coexists in two phases (gas and liquid). At critical point and in the region, the temperature and the pressure are above the critical temperature and pressure, the substance exists in a single gaseous phase. The

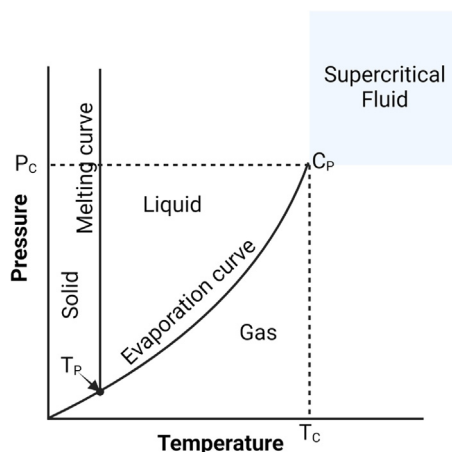


Figure 2.8 SFE phase diagram of supercritical fluid region. From Saito, M. (2013). History of supercritical fluid chromatography: Instrumental development. *Journal of Bioscience and Bioengineering*, 115(6), 590–599. <https://doi.org/10.1016/j.jbiosc.2012.12.008>.

substance in this region is called supercritical state, where the density is liquid-like while the viscosity is gas-like, and the diffusivity is in between those of a liquid and a gas (Saito, 2013).

The increasing interest in SFE over conventional extraction in food sector is due to the following advantages (Lang & Wai, 2001):

- *Favorable viscosity and diffusivity:* Supercritical fluids possess extremely lower viscosity and higher diffusivity. Thus, SCF penetrates through the porous structure of solid material more than liquid solvents and results in superior mass transfer characteristics.
- *Selective extraction of bioactive compounds:* By varying the process parameters and choosing the correct gas, it is possible to extract the selected bioactive compounds from the plant matrix.
- *Use of fresh fluid for extraction:* In SFE, fresh fluid is continuously forced through the material, which improves the extraction yield.
- *Ability to manipulate the solvating power:* The solvating power of SCF is governed by pressure, temperature, and density. Thus, the density of SCF can be modified by changing its operating pressure and/or temperature. Controlling these parameters can improve extraction efficiency.
- *Low operating temperature:* The low temperature used for SFE makes it suitable for extracting heat-labile bioactive compounds.
- *Use of cosolvent:* SC-CO₂ (i.e., supercritical CO₂), with its nonpolar characteristics, is ideal for extracting the volatile compounds. For extracting polar compounds (like polyphenols), polar cosolvent (like ethanol) is mixed with SC-CO₂ to improve its solubility.

Several solvents are available for the SFE technique including carbon dioxide, ethane, nitrous oxide, etc. (see Table 2.6 for the detailed list). SC-CO₂ is the most preferable choice in food sector because of its numerous advantages: (i) It has easily achievable critical temperature and pressure ($T_c = 31^\circ\text{C}$ and $P_c = 72\text{ atm}$), (ii) It can be easily volatilized by releasing the pressure, and provides solvent-free extract, (iii) unlike organic solvents, SC-CO₂ is protected from air and does not pose any threat to the environment, and (iv) SC-CO₂ innocuous to human health and generally recognized as safe (More et al., 2022).

However, the major disadvantage of the SFE is the extensive equipment cost when compared to the conventional extraction technique. SFE unit consists of high-pressure pump, extraction chamber, recovery chamber, and collection device. The SFE system operated continuously for

Table 2.6 Properties of supercritical fluids (da Silva et al., 2016).

Supercritical fluid	T_c (°C)	P_c (atm)	Density (g/mL)	Solubility ($\text{cal}^{-1/2} \text{cm}^{-3/2}$)
Carbon dioxide	31.2	72.9	0.470	7.5
Ethane	32.4	48.2	0.200	5.8
Ethene	10.1	50.5	0.200	5.8
Methanol	−34.4	79.9	0.272	8.9
Nitrous Oxide	36.7	71.7	0.460	7.2
n-Butene	−139.9	36.0	0.221	5.2
n-Pentane	−76.5	33.3	0.237	5.1
Sulfur hexafluoride	45.8	37.7	0.730	5.5
Water	101.1	217.6	0.322	13.5

Source: From da Silva, R. P. F. F., Rocha-Santos, T. A. P. & Duarte, A. C. (2016). Supercritical fluid extraction of bioactive compounds. *TrAC - Trends in Analytical Chemistry*, 76, 40–51. <https://doi.org/10.1016/j.trac.2015.11.013>.

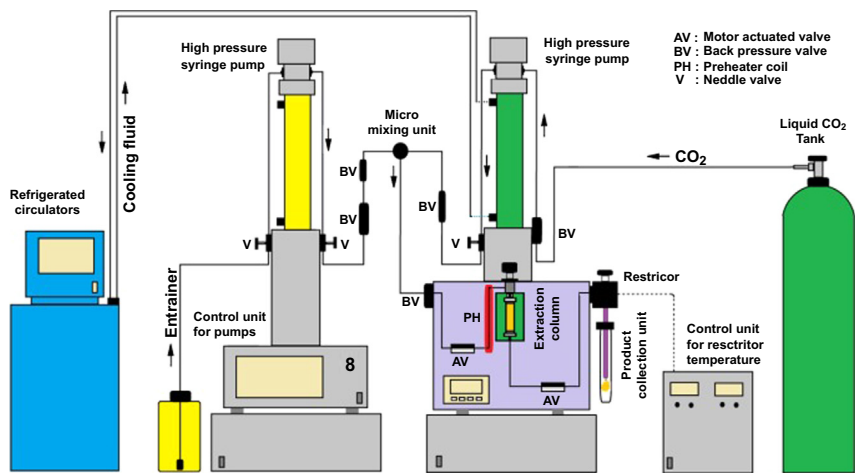


Figure 2.9 Supercritical fluid extraction instrument supercritical fluid extraction system. From Salgın, Ul., Salgın, S., Ekici, D. D. & Uludağ, G. (2016). Oil recovery in rosehip seeds from food plant waste products using supercritical CO₂ extraction. *The Journal of Supercritical Fluids*, 118, 194–202. <https://doi.org/10.1016/j.supflu.2016.08.011>.

fluid phase and batch-wise for the solid phase (Salgın et al., 2016). See Fig. 2.9 for the SFE system. SFE begins with pressuring the CO₂ above its critical pressure. In the extraction chamber, SC-CO₂ passes through the plant matrix and extracts the bioactive compounds. The bioactive from the SC-CO₂ is collected from the recovery chamber by reducing the pressure at constant temperature.

Extraction mechanism: Briefly, supercritical extraction (SCE) occurs in two stages: solubilization of bioactive compounds from the solid matrix and its separation into supercritical solvent. Similar to conventional extraction technique, the solvent flows through the packed bed, dilates the intercellular channels, and solubilizes the bioactive compounds in the plant matrix. However, in conventional extraction technique, the bioactive compound is dissolved in the solvent and processed further to isolate the bioactive compounds. In SCE, isolation is immediately achieved by reducing the pressure and/or increasing the temperature (Brunner, 1994; da Silva et al., 2016).

Extraction of bioactive compounds (like volatile compounds and essential oils) from the plant matrix follows three mass transfer mechanisms (da Silva et al., 2016; Mezzomo et al., 2009):

- *Constant extraction rate (CER):* External surface of materials covered with solute and easily accessible to solute. Transfer of bioactive compounds follows convection (see Fig. 2.10).
- *Falling extraction rate (FER):* Oil content in the external surface decreases and diffusion starts along with convection.

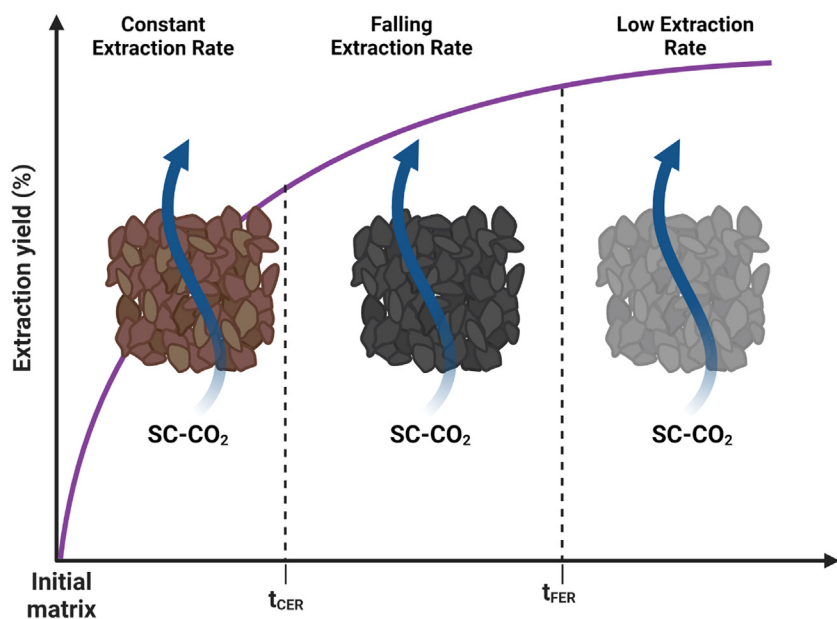


Figure 2.10 SFE Extraction period different extraction period of supercritical fluid extraction. Created with Biorender.com.

- *Low extraction rate (or diffusion-controlled)*: External layer of oils disappeared, and the mass transfer occurs by diffusion of oil from internal core.

However, the high installation cost and selected nature of SC-CO₂ in extracting small nonpolar molecules (Khaw et al., 2017) restrict the wide application of this extraction technique in the food industries.

Pulsed electric field

The application of electric fields in food processing was started in 1920s for inactivating microorganisms and enzymes from milk. Later, in mid-1960s, researchers found that nonthermal electric field can provide lethal effects on bacteria such as *Escherichia coli*, *Staphylococcus aureus*, *Micrococcus lysodeikticus*, *Sarcina lutea*, *Bacillus subtilis*, *Bacillus cereus*, *Bacillus megaterium*, and *Clostridium perfringens*, as well as effects on yeasts such as *Saccharomyces cerevisiae* and *Candida utilis* (Vega-Mercado et al., 1997). This nonthermal and low-energy PEF process was later utilized as pretreatment for conventional extraction technique and authors found PEF pretreatment increased the extraction yield (El Kantar et al., 2018).

PEF extraction technique is one of the innovative nonthermal extraction techniques, effectively utilizes the electric pulses to rupture the microbial and plant cell wall. During PEF extraction, high-voltage (kV) pulses of short time intervals (microseconds to milliseconds) were applied to the plant matrix or other material through the electrodes (Martínez et al., 2020). When passing the high electric field across the biological cells, the cell membrane becomes permeable to molecules and this behavior is referred to as “electro-permeabilization” or “electroporation.” Chemat, Rombaut, Meullemiestre, et al. (2017) reported the effectiveness of cell membrane’s electroporation depends on the process parameters (such as electric field strength, treatment time, frequency, and temperature), treatment mode (batch or continuous), configuration of treatment chamber (collinear, coaxial and parallel), physicochemical characteristics of the treated matrix (pH and conductivity), characteristics of the treated cells (size, shape, membrane, and envelope structure), and state (suspension, solid, semisolid). By varying the process parameters (electric field strength, pulse duration, and energy input), it is possible to use the PEF for the following applications (Picart-Palmade et al., 2019):

1. microbial inactivation (15–40 kV/cm to 1,000 kJ/kg),
2. sludge disintegration (10–20 kV/cm to 50–200 kJ/kg),

3. improvement of mass transfer in plant or animal cells (0.7–3.0 kV/cm to 1–20 kJ/kg),
4. reversible electro-permeabilization of biological cells for DNA transfer (0.7 kV/cm to 1–10 kJ/kg), and
5. induction of stress response (0.5–1.5 kV/cm to 0.5–5 kJ/kg).

PEF is gaining attention for the extraction of valuable bioactive compounds including vitamins, carotenoids, sucrose, proteins, inulin, and for the formulation of functional beverages. [Siddeeg et al. \(2019\)](#) developed alcoholic beverages of date palm fruits by treating the palm date fruits with PEF at three different electric strength (1.38, 2.02, and 2.92 kV/cm) for 100 μ s (treatment time). The authors found that the alcoholic beverages treated with PEF showed a significant increase in nutritional value than the untreated beverages. The amount of free amino acid, carotenoids, anthocyanin, flavonoid, and phenolic content was significantly higher in the PEF-treated beverages and the nutritional value increased with the applied electric strength $2.92 > 2.02 > 1.38$ kV/cm.

[Ahmad Shiekh et al. \(2021\)](#) also reported a significant improvement in extraction yield, total phenolic content, and antioxidant and antimicrobial properties of custard apple leaf extract. The increase in antioxidant properties of PEF-assisted leaf extract is due to the inhibition of chlorophyll and carotenoid degradation. PEF can form chlorophyll-aggregated structures by promoting cross-linking reactions and forming dimeric and oligomeric chlorophyll complexes. This chlorophyll complex protects the pyrrole ring of the chlorophyll molecule and potentially improved the chlorophyll content of the extract ([Zhang et al., 2017](#)).

Further, the energy consumption of PEF-assisted extraction typically requires between 1 and 15 kJ/kg, which is significantly lower than the energy requirement of conventional treatments like mechanical pressing (20–40 kJ/kg), enzymatic (60–100 kJ/kg) and heating (> 100 kJ/kg) ([Barba et al., 2015](#)). However, the major limitation of the PEF is high initial investment and processing cost, which restricts the wide application of PEF in food processing sector. For instance, installing a PEF unit (20–40 kV/cm, 100–1,000 kJ/k) with 5t/h capacity may require \$2–3 million ([Picart-Palmade et al., 2019](#)). The increase in capital cost will potentially increase the processing cost. [Sampedro et al. \(2013\)](#) estimated the production cost for PEF pasteurization of orange juice was 0.037 \$/L, whereas it costs only 0.015 \$/L by thermal pasteurization.

Enzyme-assisted extraction

Enzyme-assisted method is a potential (alternative) green extraction technique and is gaining attention for its efficiency, sustainability, and eco-friendly approach. Briefly, enzymes are used to disrupt the plant cell walls and intercellular components are released. Three major polysaccharides (pectin, hemicellulose, and cellulose) serve as a barrier in releasing its intercellular compounds. In EAE, these cell wall polysaccharides are broken down by pectinase, hemicellulases, xylanase, β -glucanase, feruloyl esterases, and β -glucosidase and release the bioactive compounds (Ali et al., 2022).

The major advantage of EAE is that it can work effectively under adverse conditions and had the capacity to act in synergy with other solvents and techniques. However, the major drawback of this EAE technique is the cost of enzyme, which are very expensive for industrial-scale operation and the inability of the enzymes to completely hydrolyze the plant cell walls (Marić et al., 2018). Thus, several combinations of extraction techniques with EAE were developed to improve the extraction efficiency of bioactive compounds in more environmentally friendly situations (Alavarsa-Cascales et al., 2022).

Bautista-Expósito et al. (2022) valorized the lentil hulls through the extraction of oligosaccharides and phenolic compounds using MAE, EAE, and integrated microwave and EAE techniques. The authors used seven commercial enzymes for the EAE and studied the release of oligosaccharides and phenolic compounds and their antioxidant activity. The highest oligosaccharide extractability was achieved for Pectinex Ultra Tropical (14.17 g/100 g DW), Shearzyme Plus 2X (9.79 g/100 g DW), and Viscozyme L (7.54 g/100 g DW) due to the varying enzyme activities. Extraction of oligosaccharides using Pectinex Ultra Tropical was compared with the microwave-assisted extraction and its combination (MAE + EAE). Authors found that MAE resulted in significantly higher amounts of oligosaccharides (14.17 g/100 g DW) with antioxidant activity (442 μ mol TE/g DW) than the MAE (4.65 ± 0.23 g oligosaccharide/100 g DW) and combination techniques (12.12 ± 0.81 g oligosaccharide/100 g DW). Spadoni Andreani and Karboune (2020) also noticed a significant increase in bioactive extraction (cell wall carbohydrates like pectic oligo/polysaccharides) from cranberry pomace using EAE than the MAE. The decreasing extraction yield for MAE might be attributed to the high operating temperature and microwave radiation, which causes the

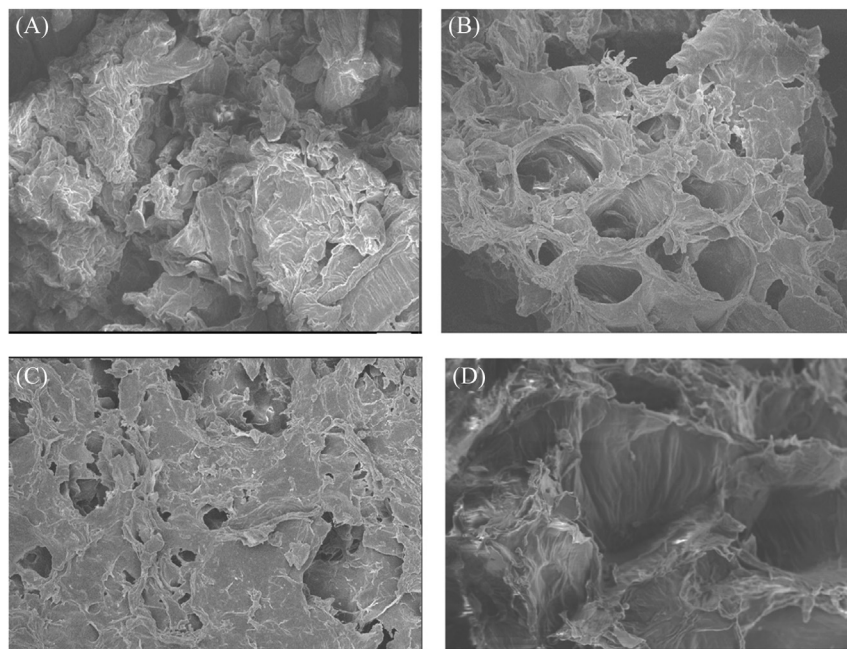


Figure 2.11 Supercritical fluid extraction figure for enzymatic treatment effect of treatment on the surface morphology of treated and untreated grapefruit insoluble dietary fiber (A) untreated sample showing tight and wrinkle structure, (B) microwave treatment, (C) enzymatic treatment, and (D) combination of microwave and enzymatic treatment. From Peng, G., Gan, J., Dong, R., Chen, Y., Xie, J., Huang, Z., Gu, Y., Huang, D. & Yu, Q. (2021). Combined microwave and enzymatic treatment improve the release of insoluble bound phenolic compounds from the grapefruit peel insoluble dietary fiber. *LWT*, 149, 111905. <https://doi.org/10.1016/j.lwt.2021.111905>.

structural degradation of phenolic compounds (Singla et al., 2022). In the case of grapefruit peel insoluble dietary fiber, the combination of MAE + MAE showed a significantly higher content of phenols 12.46 ± 0.028 GAE mg/100 g than the MAE (9.95 ± 0.04 GAE mg/100 g) and EAE (7.17 ± 0.044 GAE mg/100 g) (Peng et al., 2021). The variation in the surface morphology of the grapefruit peel insoluble dietary fiber after different treatments can be observed through the SEM images (see Fig. 2.11).

The tight and wrinkle surface in the untreated samples represents the cell wall with a tight network of cellulose, hemicellulose, and lignin compositions. During microwave and enzymatic treatment, the structure became looser with the formation of porous honeycomb-like structure,

representing the partial removal of lignin and hemicellulose. The weakening/breaking behavior of cell wall by enzymes helps in diffusing more bioactive compounds into the extracting medium.



Future trends

Extraction is the first industrial process of isolating bioactive compounds from the plant material and adding commercial value to the agro products. Traditional extraction techniques were employed for many decades for extracting bioactive compounds and treating/preventing diseases. However, the numerous drawbacks, such as time consumption, thermal stability, use of organic solvents, and toxicity, restrict the researchers to use these techniques. Novel extraction techniques, such as SFE, microwave-assisted extraction, UAE, PEF, resulted in maximum extraction yield with less operation time and lesser environmental issues. However, each extraction technique has its own drawbacks such as the installation cost for SCE, PEF, UAE, enzyme cost for EAE, change of chemical structure in MAE, etc. On the other hand, green solvents like ionic liquids, deep eutectic solvents, and natural eutectic solvents showed promising results in extracting phenolic compounds, terpenoids, and essential oils from plant materials. The potential of these green solvents in novel extraction techniques needs to be explored further.

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